Genetics, atrophy, liquid biomarkers and cognitive tests in Alzheimer's disease: an integrated and multiomics analysis pipeline to understand the neurodegenerative mechanisms.

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Abstract

- Alzheimer's disease (AD) is the most common neurodegenerative disease among the elderly (50–75%) and the causes of the sporadic form are not yet clear[1]. The genetic predisposition is relevant (60–80%)[2], but alone is not sufficent to trigger those initial processes that lead to neurodegeneration and to the slowly atrophy of the brain.

- An appropriate integration of various types of data (*e.g.,* biomedical images, GWAS/WES, cognitive tests, liquor protein expression) is pivotal to understand this phenomenon[3].

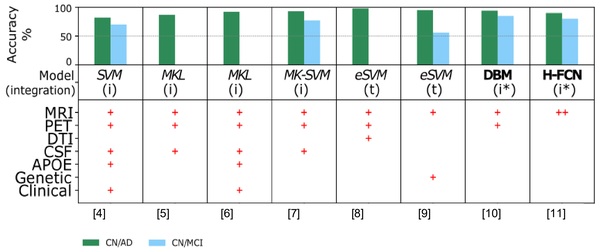
- Several pubblic datasets containing multimomics data and centered on AD (see «Public Dataset & Tools») have emerged since the beginning of ADNI in 2004.

- The objectives of this project are the following: (1) to explore and integrate these multiomics data to build a «**Merged AD Dataset**»; (2) **to develop a multikernel or deep learning pipeline** to analyzed the merged dataset (3) **to transform this analysis pipeline in a public tool** for non-data analysis experts; (3) to unveil multiomics features (*e.g.,* the gene, protein, clinical parameters and cognitive test levels) able to explain the different atrophy affecting the most relevant brain regions in the AD context. The analysis of the merged AD dataset will be lead by the use of a robust gene-SNPs panel identified in a previous study[12].

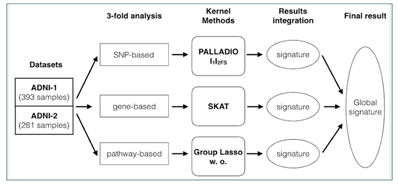
Public Dataset & Tools[13]

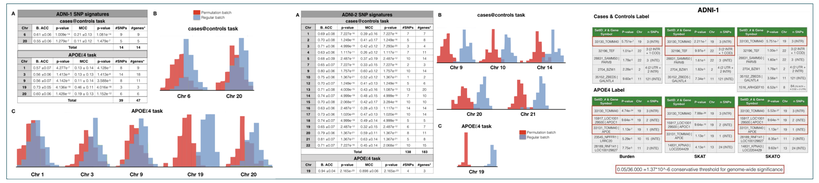
Dataset	URL
ADNI	http://adni.loni.usc.edu/
MCSA	https://www.mayo.edu/research/centers-programs/alzheimers-disease-research-center/research-activities/mayo-clinic-study-aging/for-researchers/data-sharing-resources
AIBL	https://aibl.csiro.au/
AMYPAD	https://amypad.eu/
Tools	**URL**
ALZFORUM	https://www.alzforum.org/databases
GAAIN	http://gaain.org
AData (Viewer)	https://adata.scai.fraunhofer.de

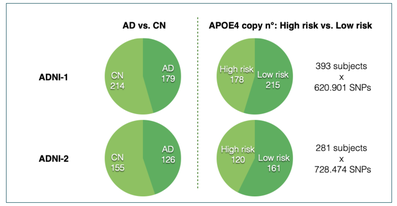
Literature on data integration approaches for AD study

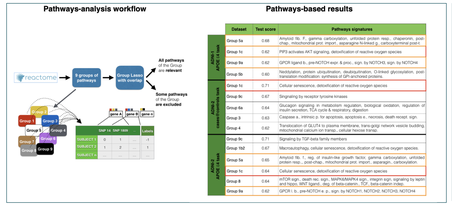


Previous study on a GWAS dataset of ADNI [12]









Conclusions

- Possible role of gene **SHLD1** (chr. 20) and its SNPs (rs6053572, rs236137 and rs1287032) in AD. SHLD1 has been **identified in both the classification tasks addressed in ADNI-1 dataset.** SHLD1 is implicated in the DNA repairing mechanism.

- In ADNI-2, **the identified SNP signature** associated to the AD risk it's very interesting from both the statistical and biological viewpoints: (i) it's classification performances are very good (the balanced accuracy is 94% and the MCC is close to 1 (that indicates perfect classification prediction) and (ii) their SNPs lie on a known linkage region associated to AD on chr. 19 (i.e.,19q13.31), but they map close to genes (i.e., LOC101928063, ZNF221 and ZNF155) not associated to AD or to other neurodegenerative disorders.

- "**Neuroactive ligand-receptor interaction**" KEGG pathway was found enriched in both the SNPs signatures identified in ADNI-1 and ADNI-2. In this pathway we found genes that encodes for glutamate receptors. GRM7 (common between ADNI-1 and ADNI-2) plays an important role as neurotransmitter in the cerebral cortex, hippocampus, and cerebellum.

- The in silico functional characterization in DISEASE highlighted several genes associated to AD or other brain disorders that also the literature characterization confirmed.

5x1000 Workflow Project

Objectives

Objective 1: identify (i) genes-SNPs signatures that correlate with atrophy starting from the genes-SNPs panel identified in previous studies[4-12] and (ii) those clinical parameters (e.g., cognitive tests, abundance protein dataset, imaging data) that correlate more strongly with atrophy.

Objective 2: test the validity of the clinical parameters identified in Objective 1 in the dataset of the Neurological clinic (MD. Pardini) of San Martino Hospital.

Objective 3: survey of the most recent and complete public dataset on AD to build a merged-multiomics dataset focused on Alzheimer's.

Objective 4: (i) develop a multikernel/deep learning pipeline to analyze the merged-multiomic AD dataset, using also the gene-SNPs and clinical parameters identified in Objective1; (ii) transform the pipeline in a open-source tool usable by non-data analysis experts.

Methods

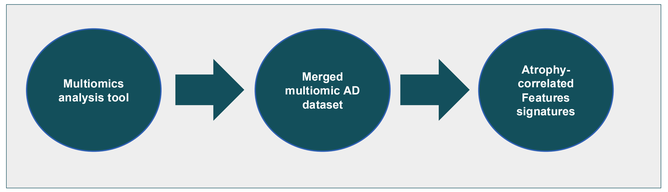
Objective 1: in public dataset AD-centered, we will evaluate the predictive performance of the gene-SNPs panel, identified in the previous study, through the use of a multikernel/deep learning pipeline, implemented in R/Python.
We will identify clusters of the genes-SNPs (of the panel) and of specific clinical parameters correlating with the atrophy (measure by FDG-PET) of the most relevant brain regions through a weighted correlation network analysis (WCNA)

Objective 2: in the dataset of the Neurological Clinic, we will validate those clinical parameters identified in objective 1 using WCNA.

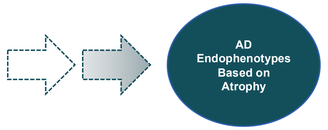
Objective 3: a fellow, under the supervision of me and PhD Reggiani, will develop a merged multiomic dataset AD-centered from a careful selection of public database.

Objective 4: Another duty of the fellow will concern the development of a user-friendly analysis tool starting from the multkernel/deep learning pipeline. This tool will be useful for the analysis of multiomics dataset such as the merged multiomics dataset built in Objective 3.

Outputs



Possible output in the future



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